MIHAELA KOSEVA

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SUMMARY

Software Engineering student at **Sofia University (FMI)** with a strong foundation in **algorithms**, **problem-solving**, and **analytical thinking**, developed at **Sofia Mathematical High School (SMG)**.

Hands-on experience in designing and developing **backend systems**, **data pipelines**, and **API-driven applications** using **Python**, **Java**, and **Clojure**, strengthened through professional experience at **ZigiWave**.

Proven ability to build **scalable** and **modular solutions**, integrate **REST APIs**, and apply **machine learning models** (including **neural networks** and **predictive analytics**) to real-world problems.

Highly motivated to contribute to **end-to-end software development**, with strong interest in **microservices architecture**, **cloud technologies**, and **enterprise-grade systems**.

PROFESSIONAL EXPERIENCE

Software Development Intern — ZigiWave, Sofia | Sep 2025 – Feb 2026

- Engineered **backend systems** and **data pipelines** using **Python**, **Java**, and **Clojure** for real-world **enterprise use cases**
- Designed and implemented **REST APIs** and **API integrations**, enabling seamless communication across **distributed systems**
- Developed **scalable**, **modular**, and **maintainable components** following **clean architecture** and **best practices**
- Contributed to **data processing workflows** and **integration pipelines**, improving **system efficiency** and **reliability**
- Actively participated in **Agile development cycles**, including **code reviews**, **team collaboration**, and **version control workflows (Git)**

SELECTED PROJECTS

Neural Network Lab (Python, TensorFlow)

- Designed, trained, and evaluated **Neural Networks (TensorFlow)** for **classification** and **function approximation (~90%+ accuracy)**
- Implemented core **algorithms** from scratch, including **Backpropagation** and **Gradient Descent**
- Applied **ML techniques** to **dynamic system modeling (SIR model, ODE-based simulations)**

Profit Predictor: End-to-End ML System (Python, Scikit-learn, Pandas, NumPy)

- Developed a full **end-to-end machine learning pipeline** for **profit forecasting** and **trend analysis**
- Conducted **EDA**, **feature engineering**, and **data preprocessing** on **large datasets**
- Implemented **Regression**, **Polynomial Regression**, **Decision Trees**, and **Time Series models** for **actionable insights**

Placebo Effect Analysis: Data Science Project (Python, Pandas, Matplotlib, NumPy)

- Performed **EDA** and **statistical analysis** on **clinical datasets (5,000+ records)**
- Conducted **hypothesis testing** and **correlation analysis (p < 0.05)**
- Improved **data quality** through **preprocessing** and created clear **data visualizations**

Deliveries Database System (SQL, Relational Databases)

- Designed **normalized relational database schema** for **reporting** and **analytics**
- Optimized complex **SQL queries**, improving **performance** by ~30%
- Enabled **KPI tracking** and **data-driven operational insights**

KEY SKILLS

Machine Learning & AI:
Neural Networks, Regression, Classification, Model Evaluation, Feature Engineering, Scikit-learn, Prompt Engineering (Basics), LLM Fundamentals

Backend Development:
Java, Python, REST APIs, API Integrations, Microservices Architecture (Concepts), Spring Boot (Basics)

Frontend Development:
JavaScript, HTML5, CSS3, REST API Integration, Responsive Web Design (Basics)

Functional Programming:
Clojure, ClojureScript, Python (Functional Paradigms)

Databases:
SQL, Relational Databases, Query Optimization, Data Modeling

Tools & Practices:
Git, Jupyter Notebook, VS Code, Agile/Scrum, Object-Oriented Programming (OOP), Clean Code Principles, Modular Architecture, Scalable Systems Design

CORE CONCEPTS

Microservices Architecture
RESTful Design Principles
Client-Server Architecture
Data Modeling & System Design

EDUCATION

Sofia University (FMI) — **Bachelor's Degree** in **Software Engineering**
Oct 2024 – Present

Sofia Mathematical High School (SMG) — **2019 – 2024 | GPA: 5.99 / 6.00**

CERTIFICATIONS & COURSES

- **Data Science** — SoftUni
- **Mathematical Concepts for Developers** — SoftUni
- **Python (Fundamentals, Advanced, OOP, Web, Frameworks)** — SoftUni

AWARDS

- **Laureate — FMI Mathematical National Competition**
- **Participant — National Mathematics & Physics Olympiads**